Rational Expressions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Key features of quadratic functions**

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of the equation $f(x) = 0$?

To find the roots of the equation $f(x) = 0$, we need to solve $x^2 - 6x + 8 = 0$. This quadratic equation can be factored. We are looking for two numbers that multiply to 8 and add up to -6. These numbers are -2 and -4. Therefore, the equation can be factored as $(x - 2)(x - 4) = 0$. Setting each factor to zero gives us the roots: $x - 2 = 0 \Rightarrow x = 2$ and $x - 4 = 0 \Rightarrow x = 4$.

2 **[Foundational]** • **Target: Solving quadratic equations by factoring**

Consider the polynomial $P(x) = x^3 - 6x^2 + kx + 12$. If $(x - 3)$ is a factor of $P(x)$, and the sum of the roots of $P(x) = 0$ is S, find the value of k. Additionally, determine the sum of the roots S given that the roots are α, β, γ.

Since $(x - 3)$ is a factor of $P(x) = x^3 - 6x^2 + kx + 12$, we know that $P(3) = 0$. Substituting $x = 3$ into the polynomial: $P(3) = (3)^3 - 6(3)^2 + k(3) + 12 = 0$ $27 - 6(9) + 3k + 12 = 0$ $27 - 54 + 3k + 12 = 0$ $-27 + 3k + 12 = 0$ $-15 + 3k = 0$ $3k = 15$ $k = 5$. For a cubic polynomial of the form $ax^3 + bx^2 + cx + d = 0$, the sum of the roots $(\alpha + \beta + \gamma)$ is given by $-\frac{b}{a}$. In our case, $P(x) = x^3 - 6x^2 + kx + 12$, so $a = 1$ and $b = -6$. The sum of the roots.

3 **[Foundational]** • **Target: Solving quadratic equations using square roots**

The rational expression $\frac{2x^2 + ax - 7}{x + 4}$ is equivalent to $2x + b + \frac{5}{x + 4}$ for all $xeq -4$, where a and b are constants. What is the value of $a + b$?

To find the values of a and b, equate the two expressions: $\frac{2x^2 + ax - 7}{x + 4} = 2x + b + \frac{5}{x + 4}$ Multiplying both sides by the common denominator $x + 4$: $2x^2 + ax - 7 = (2x + b)(x + 4) + 5$ Expand the right side: $(2x + b)(x + 4) + 5 = 2x^2 + 8x + bx + 4b + 5 = 2x^2 + (b + 8)x + (4b + 5)$ Equating coefficients of corresponding powers of x on both sides: 1. Constant terms: $-7 = 4b + 5 \Rightarrow 4b = -12 \Rightarrow b = -3$. 2. Linear coefficients: $a = b + 8 \Rightarrow a = -3 + 8 = 5$.

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4 [Foundational] • Target: Solve quadratic equations using the quadratic formula

The quadratic function f is defined by $f(x) = (x + 3)^2 - 5$. What are the coordinates of the vertex of the graph of $y = f(x)$ in the xy-plane?

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where the point (h, k) represents the vertex of the parabola. Rewriting $f(x) = (x + 3)^2 - 5$ in this standard form: $f(x) = (x - (-3))^2 + (-5)$ From this expression, we identify: - $h = -3$ - $k = -5$ Thus, the coordinates of the vertex are $(-3, -5)$. Common Misconceptions: - $(3, -5)$: Incorrectly using $+3$ directly from the expression without accounting for the negative sign in $(x - h)$. - $(-3, 5)$ or $(3, 5)$: Inverting the sign of the.

5 [Foundational] • Target: Graphing quadratic functions

The quadratic function $f(x) = ax^2 + bx + c$ has a vertex at the point $(2, -5)$. If the graph of $f(x)$ passes through the point $(4, 3)$, find the value of a.

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -5)$, the function can be written as $f(x) = a(x - 2)^2 - 5$. We are also given that the graph passes through the point $(4, 3)$. Substituting these coordinates into the equation, we get: $3 = a(4 - 2)^2 - 5$. Simplifying, we have $3 = a(2)^2 - 5$, which means $3 = 4a - 5$. Adding 5 to both sides gives $8 = 4a$. Dividing by 4, we find $a = 2$.

6 [Foundational] • Target: Transformations of quadratics

Factor the quadratic expression: $x^2 + 5x + 6$.

To factor the quadratic expression $x^2 + 5x + 6$, we need to find two numbers that multiply to 6 and add up to 5. These two numbers are 2 and 3. Therefore, the factored form is $(x + 2)(x + 3)$. Let's check the other options: - $(x + 1)(x + 6) = x^2 + 6x + x + 6 = x^2 + 7x + 6$ (Incorrect sum) - $(x - 2)(x - 3) = x^2 - 3x - 2x + 6 = x^2 - 5x + 6$ (Incorrect sum sign) - $(x - 1)(x - 6) = x^2 - 6x - x + 6 = x^2 - 7x + 6$ (Incorrect sum sign)

7 [Foundational] • Target: Key features of quadratic functions

Consider the rational expression: $\frac{x^2 - 4}{x - 2}$. If $xeq2$, what is the simplified form of this expression?

The given rational expression is $\frac{x^2 - 4}{x - 2}$. The numerator $x^2 - 4$ is a difference of squares, which can be factored as $(x - 2)(x + 2)$. Therefore, the expression can be rewritten as $\frac{(x - 2)(x + 2)}{x - 2}$. Since it is given that $xeq2$, we can cancel the common factor $(x - 2)$ from the numerator and the denominator. This leaves us with $x + 2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Solving quadratic equations by factoring

Given the polynomial $P(x) = ax^3 - 2x^2 + bx + 5$, if $P(x)$ is divisible by $(x - 1)$ and the remainder when $P(x)$ is divided by $(x + 2)$ is -9, find the value of $a + b$.

Since $P(x)$ is divisible by $(x - 1)$, by the Remainder Theorem, $P(1) = 0$. Substituting $x = 1$ into $P(x)$: $a(1)^3 - 2(1)^2 + b(1) + 5 = 0 \Rightarrow a - 2 + b + 5 = 0 \Rightarrow a + b + 3 = 0 \Rightarrow a + b = -3$. When $P(x)$ is divided by $(x + 2)$, the remainder is -9. By the Remainder Theorem, $P(-2) = -9$. Substituting $x = -2$ into $P(x)$: $a(-2)^3 - 2(-2)^2 + b(-2) + 5 = -9$. $-8a - 2(4) - 2b + 5 = -9$ $-8a - 8 - 2b + 5 = -9$ $-8a - 2b - 3 = -9$ $-8a - 2b = -6$ Dividing by -2: $4a + b = 3$. We have a s.

9 [Intermediate] • Target: Solving quadratic equations using square roots

If the expression $\frac{\sqrt[4]{x^3 \cdot \sqrt{x^k}}}{x^{-\frac{1}{4}}}$ is equivalent to x^2 for all $x > 0$, what is the value of the constant k?

To find the value of k, convert each radical expression into rational exponent form and apply exponent rules step-by-step: 1. Convert the inner radical: $\sqrt{x^k} = (x^k)^{\frac{1}{2}} = x^{\frac{k}{2}}$ 2. Combine terms inside the fourth root using the product rule $x^a \cdot x^b = x^{a+b}$: $x^3 \cdot x^{\frac{k}{2}} = x^{3+\frac{k}{2}} = x^{\frac{6+k}{2}}$ 3. Apply the fourth root using the power rule $(x^a)^b = x^{ab}$.

10 [Intermediate] • Target: Solve quadratic equations using the quadratic formula

Triangle ABC is mapped to triangle $A'B'C'$ on the coordinate plane as shown below. Which coordinate rule represents this transformation?

To determine the transformation rule, compare the coordinates of corresponding vertices of triangle ABC and triangle $A'B'C'$: - Vertex $A(1, 2)$ maps to $A'(-4, 2)$. - Vertex $B(4, 2)$ maps to $B'(-1, 2)$. - Vertex $C(2, 5)$ maps to $C'(-3, 5)$. For each vertex, the change in the x-coordinate is: $\Delta x = -4 - 1 = -5$ The y-coordinate remains unchanged: $\Delta y = 2 - 2 = 0$ This is a horizontal translation of 5 units to the left, which is represented by the transformation rule.

11 [Intermediate] • Target: Graphing quadratic functions

Let $f(x) = \sqrt{x^2 - 4x + 4}$ and $g(x) = \sqrt[3]{x^3 + 6x^2 + 12x + 8}$. If $h(x) = f(x) \cdot g(x)$, find the value of $h(-3)$.

First, simplify the expressions for $f(x)$ and $g(x)$. For $f(x) = \sqrt{x^2 - 4x + 4}$, we recognize the expression inside the square root as a perfect square trinomial: $x^2 - 4x + 4 = (x - 2)^2$. Therefore, $f(x) = \sqrt{(x - 2)^2} = |x - 2|$. For $g(x) = \sqrt[3]{x^3 + 6x^2 + 12x + 8}$, we recognize the expression inside the cube root as a perfect cube: $x^3 + 6x^2 + 12x + 8 = (x + 2)^3$. Therefore, $g(x) = \sqrt[3]{(x + 2)^3} = x + 2$. Now, we can find the expression for $h(x) = f(x) \cdot g(x)$..

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12 [Intermediate] • Target: Transformations of quadratics

Consider the function defined by $f(x) = \sqrt{ax + b}$. If the graph of $f(x)$ passes through the points $(1, 3)$ and $(5, 5)$, determine the value of a.

The function is given by $f(x) = \sqrt{ax + b}$. The graph passes through $(1, 3)$, so $f(1) = 3$. Substituting these values into the function: $3 = \sqrt{a(1) + b}$. Squaring both sides gives $9 = a + b$ (Equation 1). The graph also passes through $(5, 5)$, so $f(5) = 5$. Substituting these values into the function: $5 = \sqrt{a(5) + b}$. Squaring both sides gives $25 = 5a + b$ (Equation 2). We now have a system of two linear equations with two variables: $a + b = 9$ and $5a + b = 25$. Subtract Equation 1 from Equ.

13 [Intermediate] • Target: Key features of quadratic functions

In the figure below, triangle ABC is a right triangle with right angle at C, and altitude CD is drawn to hypotenuse AB. The length of altitude CD is 6, $AD = x$, and $DB = x + 5$. What is the length of hypotenuse AB?

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $CD^2 = AD \cdot DB$ Substitute the given values into the equation: $6^2 = x(x + 5)$ $36 = x^2 + 5x$ Rearrange into standard quadratic form: $x^2 + 5x - 36 = 0$ Factor the quadratic equation: $(x + 9)(x - 4) = 0$ This gives two solutions: $x = -9$ and $x = 4$. Because $AD = x$ represents a geometric length, it must be positive. Thus, we reject the extraneous.

14 [Intermediate] • Target: Solving quadratic equations by factoring

The graph of the function $y = a \cdot b^x$, where $ae0$ and $b > 0, be1$, passes through the points $(1, 6)$ and $(3, 54)$. Determine the exact value of a and the exact value of b.

Let the function be $y = a \cdot b^x$. Since the graph passes through $(1, 6)$, we have $6 = a \cdot b^1$ (Equation 1). Since the graph also passes through $(3, 54)$, we have $54 = a \cdot b^3$ (Equation 2). From Equation 1, we can express a in terms of b: $a = \frac{6}{b}$. Substitute this expression for a into Equation 2: $54 = \left(\frac{6}{b}\right) \cdot b^3$ $54 = 6 \cdot b^2$ Divide both sides by 6: $b^2 = \frac{54}{6}$ $b^2 = 9$ Since $b > 0$, we take the positive square root: $b = 3$. Now substitute.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging]** • **Target: Solving quadratic equations using square roots** • **Cross-Lesson: 5.01 Exponential & Logarithmic Functions**

Consider a line that passes through the point $(2, 5)$ and has a slope of -3. If the equation of this line is written in the form $y = mx + c$, and it is also known that the point $(k, -7)$ lies on this line, determine the value of k.

The equation of a line with slope m passing through a point (x_1, y_1) can be found using the point-slope form: $y - y_1 = m(x - x_1)$. Given the point $(2, 5)$ and slope $m = -3$, we have: $y - 5 = -3(x - 2)$ Expand the equation: $y - 5 = -3x + 6$ Rearrange into the form $y = mx + c$: $y = -3x + 6 + 5$ $y = -3x + 11$ So, the equation of the line is $y = -3x + 11$. The value of c is 11. We are also given that the point $(k, -7)$ lies on this line. Substitute these coordinates into the equation.

16 **[Challenging]** • **Target: Solve quadratic equations using the quadratic formula** • **Cross-Lesson: 6.01 Systems & Inequalities**

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(0, 3)$ and $(2, 75)$. Determine the value of a.

The function is given by $f(x) = a \cdot b^x$. \\ We are given that the graph passes through $(0, 3)$. Substituting these coordinates into the function: \\ $3 = a \cdot b^0$\\ Since $b^0 = 1$ (for any non-zero b), we have: \\ $3 = a \cdot 1$\\ $a = 3$. \\ Now, we can use the second point $(2, 75)$ to find b. \\ $75 = a \cdot b^2$\\ Substitute the value of $a = 3$: \\ $75 = 3 \cdot b^2$\\ Divide both sides by 3: \\ $25 = b^2$\\ Taking the square root of both sides, we get $b = 5$ (assuming $b > 0$ for typical expo.

17 **[Challenging]** • **Target: Graphing quadratic functions** • **Cross-Lesson: 7.01 Sequences & Series**

Consider a line passing through the points $A(2, k)$ and $B(6, 10)$. If the slope of the line segment AB is m and $m = \frac{1}{2}$, find the value of k.

The slope of a line passing through two points (x_1, y_1) and (x_2, y_2) is given by the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Given the points $A(2, k)$ and $B(6, 10)$, we have $x_1 = 2$, $y_1 = k$, $x_2 = 6$, and $y_2 = 10$. The slope is given as $m = \frac{1}{2}$. Substituting the values into the slope formula: $\frac{1}{2} = \frac{10 - k}{6 - 2}$ $\frac{1}{2} = \frac{10 - k}{4}$ To solve for k, we can cross-multiply: $1 \times 4 = 2 \times (10 - k)$ $4 = 20 - 2k$ Now, isolate the term with k.

18 **[Challenging]** • **Target: Transformations of quadratics** • **Cross-Lesson: 8.01 Factoring Polynomials**

The graph of the function $f(x) = \sqrt{ax + b}$ passes through the points $(1, 3)$ and $(5, 5)$. Find the value of a and the value of b.

Given the function $f(x) = \sqrt{ax + b}$, and it passes through the points $(1, 3)$ and $(5, 5)$. We can set up a system of equations: 1. For point $(1, 3)$: $3 = \sqrt{a(1) + b} \Rightarrow 3 = \sqrt{a + b}$. Squaring both sides gives $9 = a + b$. (Equation 1) 2. For point $(5, 5)$: $5 = \sqrt{a(5) + b} \Rightarrow 5 = \sqrt{5a + b}$. Squaring both sides gives $25 = 5a + b$. (Equation 2) Now we solve the system of linear equations: Equation 2 - Equation 1: $(5a + b) - (a + b) = 25 - 9$ $4a = 16$ $a = 4$ Substitute the value of a.

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19 [Challenging] • Target: Key features of quadratic functions • Cross-Lesson: 9.01 Radicals & Exponents

The graph of the exponential function $f(x) = a \cdot b^x + c$ passes through the points $(0, 3)$ and $(1, 7)$. If the horizontal asymptote of the graph is $y = 1$, find the value of a and b.

The general form of the exponential function is $f(x) = a \cdot b^x + c$. We are given that the horizontal asymptote is $y = 1$, which means $c = 1$. So, the function becomes $f(x) = a \cdot b^x + 1$. ; The graph passes through $(0, 3)$. Substituting these coordinates into the function: ;
$3 = a \cdot b^0 + 1; 3 = a \cdot 1 + 1; 3 = a + 1; a = 2$. ; Now, the function is $f(x) = 2 \cdot b^x + 1$. ; The graph also passes through $(1, 7)$. Substituting these coordinates into the function.

20 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 1.01 Linear Equations & Lines

Let $P(x)$ be a polynomial such that $P(x^2 + 1) = x^4 + 3x^2 + 2$. If $P(x)$ can be expressed in the form $ax^2 + bx + c$, find the value of b.

We are given the equation $P(x^2 + 1) = x^4 + 3x^2 + 2$. We are also told that $P(x)$ is a quadratic polynomial of the form $ax^2 + bx + c$. Let $y = x^2 + 1$. Then $x^2 = y - 1$. Substituting this into the expression for $P(x^2 + 1)$, we get: $P(y) = (y - 1)^2 + 3(y - 1) + 2$ $P(y) = (y^2 - 2y + 1) + (3y - 3) + 2$
$P(y) = y^2 + (-2y + 3y) + (1 - 3 + 2)$ $P(y) = y^2 + y + 0$ Thus, $P(y) = y^2 + y$. Replacing y with x, we have $P(x) = x^2 + x$. Comparing this to the form $P(x) = ax^2 + bx + c$, we find that $a = 1$, $b = 1$, an.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Key features of quadratic functions

For the model $f(x) = 2x + 6$ in key features of quadratic functions, evaluate $f(2)$.

a. 10 ✓

b. 12

c. 4

d. 14

Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

22 [Foundational] • Target: Key features of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

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23 [Foundational] • Target: Key features of quadratic functions

For the model $f(x) = 2x + 6$ in key features of quadratic functions, evaluate $f(5)$.

a. 16 ✓

b. 18

c. 10

d. 20

Substitute $x = 5$: $f(5) = 2(5) + 6 = 16$.

24 [Foundational] • Target: Solving quadratic equations by factoring

For the model $f(x) = 3x + 7$ in solving quadratic equations by factoring, evaluate $f(2)$.

a. 13 ✓

b. 16

c. 6

d. 17

Substitute $x = 2$: $f(2) = 3(2) + 7 = 13$.

25 [Foundational] • Target: Solving quadratic equations by factoring

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Solving quadratic equations by factoring

For the model $f(x) = 3x + 7$ in solving quadratic equations by factoring, evaluate $f(5)$.

a. 22 ✓

b. 25

c. 15

d. 26

Substitute $x = 5$: $f(5) = 3(5) + 7 = 22$.

27 [Foundational] • Target: Solving quadratic equations using square roots

For the model $f(x) = 4x + 8$ in solving quadratic equations using square roots, evaluate $f(2)$.

a. 16 ✓

b. 20

c. 8

d. 20

Substitute $x = 2$: $f(2) = 4(2) + 8 = 16$.

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28 [Foundational] • Target: Solving quadratic equations using square roots

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Key features of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Key features of quadratic functions

A data set recorded for key features of quadratic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

31 [Intermediate] • Target: Key features of quadratic functions

For the model $f(x) = 2x + 6$ in key features of quadratic functions, evaluate $f(4)$.

a. 14 ✓

b. 16

c. 8

d. 18

Substitute $x = 4$: $f(4) = 2(4) + 6 = 14$.

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32 [Intermediate] • Target: Key features of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

33 [Intermediate] • Target: Solving quadratic equations by factoring

A data set recorded for solving quadratic equations by factoring consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

34 [Intermediate] • Target: Solving quadratic equations by factoring

For the model $f(x) = 3x + 7$ in solving quadratic equations by factoring, evaluate $f(3)$.

a. 16 ✓

b. 19

c. 9

d. 20

Substitute $x = 3$: $f(3) = 3(3) + 7 = 16$.

35 [Intermediate] • Target: Solving quadratic equations by factoring

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

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36 [Intermediate] • Target: Solving quadratic equations using square roots

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

37 [Intermediate] • Target: Solving quadratic equations using square roots

For the model $f(x) = 4x + 8$ in solving quadratic equations using square roots, evaluate $f(3)$.

a. 20 ✓

b. 24

c. 12

d. 24

Substitute $x = 3$: $f(3) = 4(3) + 8 = 20$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving quadratic equations using square roots • Cross-Lesson: 8.01 Factoring Polynomials

For the model $f(x) = 4x + 8$ in solving quadratic equations using square roots, evaluate $f(4)$.

a. 24 ✓

b. 28

c. 16

d. 28

Substitute $x = 4$: $f(4) = 4(4) + 8 = 24$.

39 [Challenging] • Target: Solving quadratic equations using square roots • Cross-Lesson: 9.01 Radicals & Exponents

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

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40 [Challenging] • Target: Solve quadratic equations using the quadratic formula • Cross-Lesson: 1.01 Linear Equations & Lines

A data set recorded for solve quadratic equations using the quadratic formula consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

41 [Challenging] • Target: Solve quadratic equations using the quadratic formula • Cross-Lesson: 11.01 Statistics & Data Analysis

For the model $f(x) = 5x + 9$ in solve quadratic equations using the quadratic formula, evaluate $f(4)$.

a. 29 ✓

b. 34

c. 20

d. 33

Substitute $x = 4$: $f(4) = 5(4) + 9 = 29$.

42 [Challenging] • Target: Solve quadratic equations using the quadratic formula • Cross-Lesson: 2.01 Quadratic Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

43 [Challenging] • Target: Graphing quadratic functions • Cross-Lesson: 3.01 Exponents & Polynomials

A data set recorded for graphing quadratic functions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

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44 [Challenging] • Target: Graphing quadratic functions • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 6x + 3$ in graphing quadratic functions, evaluate $f(4)$.

a. 27 ✓

b. 33

c. 24

d. 31

Substitute $x = 4$: $f(4) = 6(4) + 3 = 27$.

45 [Challenging] • Target: Transformations of quadratics • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 2x + 4$ in transformations of quadratics, evaluate $f(3)$.

a. 10 ✓

b. 12

c. 6

d. 14

Substitute $x = 3$: $f(3) = 2(3) + 4 = 10$.

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